

Alicia Pankka

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Software Engineer with 5+ years of experience in game, mobile, and full-stack, specialized in GIS visualization and digital twin integration. Proven team player in a fast-paced work environment, with mentorship under high-pressure conditions. Experienced in scalable data pipelines, API design, and system architecture for production, including geospatial datasets.

Education

MSc in Computer Science (Distributed Systems), University of Tartu, Estonia (2022-2024)

BSc in Science and Technology (Game Art & Design), Thammasat University, Thailand (2015-2019)

Professional Experience

XR Developer | Helsinki XR Center (HXRC) (Jan 2025 - Present)

- Initiate and develop several RDI related to XR and geospatial data, digital twin projects.
- Organize development tasks, estimate difficulty, and distribute multiple simultaneous tasks.
- Built tool pipelines and transformation logic across projects to facilitate the workflows.
- Define technical architecture and contribute to a scalable, modular, and maintainable setup.
- Collaborate with teams and partner companies throughout development cycles.
- Lead and supervise juniors and team members in high-pressure project environments.
- Organize in-house showroom visits and workshops, after-work sessions, and larger events.

Frontend: Unity3D Developer | [Wildchain](#) (Digital Wild Animals) (Jan 2022 - Oct 2022)

- Develop an authentication system and cooperate with a backend team
- Design and create UI architecture and logic
- Display data from API to GUI and in-game sequences using UniRx
- Cooperate with the team designer to visualise features correctly
- Fix bugs before release for senior developers
- Polish in-game graphic system

Unity3D Developer | Music Ed-tech, BNK Musicmall Co.,Ltd - [Notero](#) (June 2021 - Dec 2021)

- Connect the game and the MIDI Keyboard using the MIDI Unified plugin
- Initialise game logic and classes
- Maintain up-to-date algorithms based on UML from a senior developer
- Discuss with other developers how to link data between game-sequences
- Create responsive prefabs: UI, the visual piano and raindrops

Unity3D Developer | [Ragnarok: Battle Academy](#), [Electronic Extreme](#) (Jan 2020 - May 2021)

- Manage and develop 3D asset pipelines and automation.
- Develop inventory and shop features through Unity Editor and frontend.
- Implement an authentication and experience with Firebase.
- Collaborate with UI artists to integrate UI into Unity.

Relevant Work Experience (Internships)

UE5 Integration Engineer ([Flair](#)), Artineering (Mar 2024 - May 2024)

- Prototype the new version in C++ UE5, focusing on making stylised HLSL shaders.
- Work closely with the team lead to adapt the original Flair tools to be compatible with UE5.
- Implement documentation to assist team members in understanding the current architecture and utilizing the new tools effectively.

C++ and Linux Administration ([LizardFs](#), SaunaFS), Leil Storage (Diaway) (July 2023 - Aug 2023)

- Develop Dockerfiles to containerize LizardFS services for development and test environments.
- Automate the Debian build packages pipeline and deployment workflows using Bash scripting.
- Engaging in the DevOps process, working with tools like Docker and integration tests.
- Contribute to client-server interactions by working with RPC and user session management.
- Manage user sessions on the admin side and communicate with the server using LizardFs RPC.
- Explore its chunk server and modified source code within a distributed filesystem codebase.

Professional Projects

[CosoraAtlas](#) — Full-Stack Cosplay GIS Map (Nordic/Baltic)

Frontend (Client)

- Build an interactive GIS web app that maps anime conventions and cosplay-friendly venues.
- Render a 3D vector map with MapLibre GL.
- Filter venues by city, click to fly to a location, and zoom in for detail cards.
- Implement a navigable event calendar, a light/dark theme, and
- Implement an admin dashboard for creating, editing, and deleting events.
- Design the app to run standalone with sample data, then wire it to the backend.

Backend (Server)

- Architect as a monorepo of independently deployable Cloudflare Workers microservices.
- Build REST APIs with Hono, backed by a SQLite (D1), covering full CRUD.
- Defined a shared, schema-validated data contract (with Zod) to keep services consistent.
- Create a proxy service that gets and caches official geometry from external mapping services.

[AAXLP VR Factory Simulation](#)

- Implement a factory system in Unity with a runtime workstation and tool creation
- Design a data export model for user state tracking and persistence (JSON-based)
- Initiate a structured architecture following the SOLID principle.
- Lead and supervise junior contributors under high-pressure development conditions
- Collaborate with the client during optimization and testing phases
- Integrate the Unity VR-based project into the cloud platform (VIROO)

[ENPA \(Kaupunkidata – City Data Integration\)](#)

- Integrate geospatial datasets into XR digital twin systems using API-driven workflows.
- Work with Mapbox SDK for spatial visualization and interaction in WebGL environments.
- Develop data pipelines for aggregating environmental and energy datasets
- Build and develop data pipelines for aggregating energy data (daily, weekly, monthly).

Metadata to Metaverse

- Work with the Mapbox SDK for 3D city visualization and interaction in Meta Quest 3.
- Integrate Mapbox-generated 3D data into in-house infrastructure, including geocoding, to local function within XR environments.

VLL (Virtual Learning Platform)

- Develop **full-stack** features across backend (NestJS), client applications (student/teacher), and shared in-house SDK systems.
- Implement classroom management systems and APIs supporting interactions.
- Integrate multiplayer functionality using Photon Fusion for live classroom scenarios.
- Build FSM NPC systems and integrate with the Convai conversation tool, connecting dialogue with state machines and contextual behavior.

Other projects: [KebabLala](#), [WaterSort](#) with shaders, [FlairUE5](#)

CORE SKILLS

Backend

- Python, FastAPI, Flask, Django
- REST API design, async services
- UniRx (Reactive Extensions for Unity)
- JSON serialization, data processing, ETL logic

Mobile / Platform (Familiarity)

- Android fundamentals
- Mobile SDK concepts

Databases

- Relational DB: PostgreSQL and MySQL

Frontend

- React / Vue.js
- Component-based UI architecture, state management, API-driven UIs
- Dotween, Convai, Animation pipelines

DevOps / Cloud

- Docker, Docker Compose, Kubernetes, Linux environments(Ubuntu)
- CI/CD: GitHub Actions, GitLab CI
- Basic AWS knowledge (S3, EC2, Lambda concepts)

AI / LLM EXPERIENCE (RAG-related)

- Integrated LLM-driven workflows (ChatGPT) into applications for prompt automation, dialogue systems, and contextual interactions.
- Worked with conversational AI systems (Convai) in Unity, connecting NPC behavior, state transitions, and real-time interactions to dynamic AI-generated responses.

CERTIFICATIONS

- TESOL Certification (communication, teaching, mentoring)
- AWS & DevOps self-study coursework (containers, CI/CD, monitoring)

LANGUAGES

- English (C1 – professional)
- Finnish (A2 – basic working)
- Estonian (A1)